Nurliyana Farhanah binti Mohd Zaki (24006257)

Assignment 2

Questions

1. Why does BFS ignore the distances stored in the graph even though they are available to it?

   BFS is the algorithm for uninformed search where it does not know any information like the cost and heuristic value to reach the goal. BFS explores airports in the order they enter the queue(first in first out)

2. Which two lines change when bfs is turned into dfs, and which of them makes DFS order dependent?
   BFS :

   frontier = deque([start])

   state = frontier.popleft()

   DFS:
   frontier = [start]

   state = frontier.pop()

   There are two main changes from a queue to a list used as a stack

   the removal operation change from popleft() to pop(). A queue removes the oldest item while a stack removes the newest item.

3. UCS tests for the goal when a state is popped from the heap. What could go wrong if it tested when the state was first generated?

   Because UCS prioritises the lowest accumulated cost. For example, there are route A with 10000 km and route B with 8000Km. Suppose the algorithm discovers route A first. If it immediately stops when it first discovers the destination, it might return route A without finding route B.

4. Lab1reports *expanded* and *runtime_ms*. Which one should you use to compare BFS and UCS with a classmate's run on a different laptop, and why?

   expanded is how many states had their neighbours generated while runtime_ms is how long the algorithm took to execute . Expanded is much fairer because it is less affected by hardware speed

5. Which quantity does GBFS ignore that A⋆ uses, and which failure in Lab2 does that explain?

   GBFS ignores g which is the cost from current state to the next state. GBFS may choose airport that appears closer to the destination using heuristic value, but the reaching cost may be expensive. In Lab 2, For example, for KUL -> KEF, GBFS found a route costing 12,500.5 km, while A* found shorter route costing 11,461.3 km. GBFS prioritises airports that appear closer to the destination without considering the total cost already travelled.

6. Frontier states *X* and *Y* have *(g, h)* = *(8,2)* and *(3,6)*. Which does UCS expand first? GBFS? A⋆?

UCS chooses Y because it compares the lower g

GBFS chooses X because it compares the lower h

A* chooses Y because when calculate g + h, Y has lower value

7. Why must *h* be expressed in the same units as the edge costs?

   heuristic and edge cost must use compatible units because A* calculates evaluation function using f(n) = g(n) + h(n) where g(n) is the actual cost from starting point to current point while h(n) is estimated cost from current state to the goal. if it uses different units, the heuristic could disproportionately influence the search.

8. Lab2 computes $h\star$ by running Dijkstra on the reversed graph from the goal. Why reversed?

   graph is reversed so that djikstra algorithm can calculate shortest distance from every airport to the goal in single run. By reversing the edges and starting from the goal, the shortest distance from the goal to each airport in reversed graph is the same as the shortest distance from the airport to the goal

9. Weighted A$\star$ with $w = 1.5$ changed the cost for KUL→KEF but not for SIN→EZE. What property of the SIN→EZE route explains this?

   for SIN -> EZE, weighted A* still finds the shortest route SIN -> JNB -> GRU ->EZE, with total distance of 17,817.36 km. even when heuristic weight increases to 1.5 . This is because increasing the heuristic weight does not change the route selected in this case, although the algorithm explores fewer states

Create word file and answer all the above questions. Put this in your Git Hub repo and share with me your GitHub link for AI folder as well as this file link.

Sharing GitHub Repository link

Once you have pushed your Word file to GitHub, you can easily share the repository link with me.

Here is exactly how to find and share the correct link:

1. Copy the Repository URL

1. Go to GitHub and navigate to your repository page.
2. Look at your browser's address bar. The URL should look exactly like this: https://github.com
3. **Copy** that link.

2. Verify Your Repository Visibility (Crucial Step)

Before sending the link, make sure your TA can actually open it:

- **If it is a Public Repository:** Your TA will be able to see it immediately.
- **If it is a Private Repository:** Your TA will get a **404 Error (Page Not Found)** unless you explicitly invite them. So better to have public one.